**CARD 1 OF 12** ENGINEERED HANGERS **QUESTION**

You don't have to follow Chapter 17's hanger rules if a **PE certifies** the hangers instead. What does that certification have to cover?

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**CARD 2 OF 12** NON-SYSTEM LOADS **QUESTION**

The electrician wants to zip-tie a conduit run to your branch line. **Allowed?**

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**CARD 3 OF 12** SHARED SUPPORTS **QUESTION**

A PE is designing a **shared support structure** for sprinkler pipe plus plumbing and ductwork. What two ways can it be sized?

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**CARD 4 OF 12** LISTING **QUESTION**

Which hanger components have to be **listed**, and which ones get a pass?

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**CARD 2 OF 12** NON-SYSTEM LOADS **ANSWER**

**No.** Sprinkler pipe and sprinkler hangers can't carry anything that isn't part of the sprinkler system.

**The legal way to share:** a **shared support structure**, like a trapeze rack that carries sprinkler pipe and other trades together. It has to be **certified by a PE**.

**On the job:** conduit, cable tray, signs, ductwork, light fixtures. None of them hang from your pipe.

*NFPA 13 (2019) · Ch. 17*

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**CARD 1 OF 12** ENGINEERED HANGERS **ANSWER**

**Five things:**

- Each support point holds **5× the weight of the water-filled pipe + 250 lb**
- The support points are adequate for the system
- Hanger spacing stays **within the spacing tables** for that pipe type
- Components are **ferrous** (iron or steel)
- **Calculations** showing stresses and safety factors, when the reviewer asks for them

*NFPA 13 (2019) · Ch. 17*

**Trap:** the PE route doesn't let you stretch spacing. The spacing tables still cap it.

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**CARD 4 OF 12** LISTING **ANSWER**

**Must be listed:** anything that attaches **directly** to the pipe, the building structure, or a storage rack.

**Don't have to be listed:**

- **Mild steel** hanger rods, and hangers bent from mild steel rod
- Fasteners used per the concrete, steel and wood fastener rules in Ch. 17

**Other fasteners** are OK when they're part of a hanger assembly that's been tested, listed and installed per its listing.

*NFPA 13 (2019) · Ch. 17*

**Trap:** "It's just a rod." Plain mild steel all-thread is exempt. The ring, the clamp and the beam attachment are not.

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**CARD 3 OF 12** SHARED SUPPORTS **ANSWER**

**Option 1:** 5× the weight of the water-filled sprinkler pipe **and** everything else on it, plus 250 lb (based on allowable ultimate stress).

**Option 2:** 5× the water-filled sprinkler pipe plus 250 lb, **and** 1.5× the weight of the other systems.

**Two limits:** the building structure itself doesn't count as a shared support structure, and systems that don't get along (vibration, thermal expansion) can't share one.

*NFPA 13 (2019) · Ch. 17*

**On the job:** a hot water line that grows and shrinks all day is a bad partner on a shared trapeze.

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*sheet 1 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back*

**CARD 5 OF 12** MATERIAL **QUESTION**

What do hangers have to be **made of**, and what's the exception?

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**CARD 6 OF 12** HOLES AS HANGERS **QUESTION**

Your pipe runs **through holes** in solid wood joists. Do the holes count as hangers?

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**CARD 7 OF 12** ROD SIZE **QUESTION**

What's the minimum **hanger rod** diameter for a **4 in.**, a **6 in.** and a **10 in.** pipe?

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**CARD 8 OF 12** STEEP BRANCH LINES **QUESTION**

The branch lines follow a roof pitched **6 in 12**. What extra do the branch line hanger rods need?

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**CARD 6 OF 12** HOLES AS HANGERS **ANSWER**

**They can.** Holes through solid structural members can serve as hangers **if**:
• the **building code** allows the holes, **and**
• the normal **hanger spacing and support rules** are still met.

**On the job:** a branch line running through bored joists in a house or a wood-frame building. Each hole supports the pipe like a ring would.

*NFPA 13 (2019) · Ch. 17*

**Trap:** the holes don't buy you longer spacing. If the joists are too far apart, you still add hangers.



**CARD 5 OF 12** MATERIAL **ANSWER**

**Ferrous metal** (iron or steel).

**Nonferrous is OK only if** it's been **fire tested** for the hazard, **listed** for that use, and meets the rest of the hanger rules.

**Why ferrous:** in a fire, the hangers have to keep holding the pipe up while the sprinklers flow. Low-melting metals and plastics can let go.

*NFPA 13 (2019) · Ch. 17*

**Trap:** a plastic or aluminum hanger from another trade's catalog. Unless it's listed for sprinkler use, it doesn't belong on your pipe.



**CARD 8 OF 12** STEEP BRANCH LINES **ANSWER**

**Something to take the sideways load off them.** At 6 in 12 or steeper, pick one:
• A **second hanger**, in addition to the required main hangers
• **Lateral sway braces** on the mains
• Branch line hangers with an **articulating** (swiveling) structural attachment
• Another **equivalent** way to support the branch line

*NFPA 13 (2019) · Ch. 17*

**Why:** on a steep slope, the weight of the pipe pushes downhill. A plain vertical rod gets bent sideways at the top.



**CARD 7 OF 12** ROD SIZE **ANSWER**

**4 in. pipe:** 3/8 in. rod
**6 in. pipe:** 1/2 in. rod
**10 in. pipe:** 5/8 in. rod

**The whole table:** up to 4 in. → 3/8 · 5, 6 and 8 in. → 1/2 · 10 in. → 5/8 · 12 in. → 3/4

The rod also has to be the size the **hanger assembly** was approved with.

*NFPA 13 (2019) · Ch. 17 (rod size table)*

**The exception:** a smaller rod is OK only when the hanger assembly is **tested and listed** that way, and used within its listed pipe sizes.



*sheet 2 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back*

**CARD 9 OF 12** U-HOOKS **QUESTION**

Minimum rod material for a **U-hook** on a **2 in.** pipe? On a **4 in.**?

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**CARD 10 OF 12** EYE RODS **QUESTION**

You're using **eye rods** into a wood beam. Besides rod size, what hardware does the code require?

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**CARD 11 OF 12** THREADED ROD **QUESTION**

The rod is a little long and the hanger is out of line. Can you **bend the threaded section** to make it work?

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**CARD 12 OF 12** SEISMIC LINK **QUESTION**

The job is in an area where the system has to be **protected against earthquakes**. Are Chapter 17's hanger rules enough?

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**CARD 10 OF 12** EYE RODS **ANSWER**

**Lock washers** on every eye rod, so it can't shift sideways.
**Into wood:** also a **large flat washer** bearing right against the wood, in addition to the lock washer.

**Rod size:** up to 4 in. pipe → 3/8 · 5 and 6 in. → 1/2 · 8 in. → 3/4 with a bent eye, or 1/2 with a welded eye.

*NFPA 13 (2019) · Ch. 17 (eye rod table)*

**Why the big washer:** a small washer sinks into wood under load. The large one spreads it out.



**CARD 9 OF 12** U-HOOKS **ANSWER**

**2 in. pipe:** 5/16 in.
**4 in. pipe:** 3/8 in.

**The whole table:** up to 2 in. → 5/16 · 2½ to 6 in. → 3/8 · 8 in. → 1/2

*NFPA 13 (2019) · Ch. 17 (U-hook table)*

**Trap:** mixing up the U-hook table with the hanger rod table. They only match from 2½ to 4 in. At 2 in. and smaller, and at 5 to 8 in., the numbers are different.



**CARD 12 OF 12** SEISMIC LINK **ANSWER**

**No. The hangers also have to meet the seismic hanger rules in Chapter 18.**

Chapter 17 covers holding the pipe **up**. Chapter 18 adds what keeps the hangers attached when the building **shakes**.

**Example:** on a seismic job, a **C-type beam clamp** needs a **restraining strap**, or a device specifically listed to hold the clamp to the steel.

*NFPA 13 (2019) · Ch. 17 · Ch. 18*

**Trap:** treating seismic as only sway braces. It also changes what's allowed for the hangers themselves.



**CARD 11 OF 12** THREADED ROD **ANSWER**

**No.** The threaded part of a rod can't be bent or formed.

**Why:** threads are the thinnest part of the rod. Bending them cracks the root of the thread and weakens the rod right where it carries the load.

**Fix it right:** cut the rod, move the attachment, or use a fitting that lines up.

*NFPA 13 (2019) · Ch. 17*